

A CYBER SECURITY JOB SIMULATION REPORT

Deloitte – Forage Virtual Experience

Domain: Cyber Security

Submitted to

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

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Submitted By

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Experience Details

Organisation: Deloitte

Platform: Forage

Experience: Deloitte Cyber Job Simulation

Completion Date: September 20, 2026

Academic Year 2026

CANDIDATE'S DECLARATION

I, **Harsh Raj**, hereby declare that the report titled “**Cyber Security Job Simulation – Deloitte**” has been prepared by me as part of my academic learning and documentation of the Cyber Security Job Simulation completed through Forage.

The report presents the learning, practical activities, investigation process and observations gained during the simulation. The primary case study involved investigating a simulated cybersecurity incident concerning **Daikibo Industrials**, where web activity logs were examined to identify potentially suspicious user activity.

I further declare that the information presented in this report has been prepared based on the materials, task instructions, screenshots, certificate and log data associated with the virtual experience. The report is intended for academic submission and accurately represents the nature of the simulation as a virtual job experience rather than employment or an internship with Deloitte.

I shall be responsible for maintaining academic integrity and for ensuring that the submitted material is appropriately represented.

Signature: _____

Student Name: Harsh Raj

Roll No.: 25SCS1003003055

Semester: 3rd Semester

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me in completing this Cyber Security Job Simulation.

I am thankful to **Deloitte and Forage** for providing an opportunity to experience a practical cybersecurity investigation in a simulated professional environment. The simulation helped me understand how cybersecurity professionals can approach an incident, examine web activity logs and identify suspicious patterns.

I am particularly grateful for the practical exposure provided through the **Daikibo Industrials** case study. The investigation required careful examination of web requests, authentication activity and API request patterns, which helped me connect theoretical cybersecurity concepts with practical analysis.

I would also like to thank **IILM University** and the School of Computer Science and Engineering for providing academic support and encouraging students to participate in practical learning experiences.

Finally, I express my gratitude to my teachers, family and well-wishers for their encouragement and support throughout my learning journey.

Signature: _____

Student Name: Harsh Raj

Roll No.: 25SCS1003003055

CERTIFICATE OF COMPLETION

The certificate issued through Forage confirms that **Harsh Raj** completed the **Deloitte Cyber Job Simulation** on **September 20, 2026**. The certificate states that practical tasks were completed in the field of **Cyber Security** and identifies Deloitte as the organisation associated with the simulation. It was issued by Forage.

Deloitte.

Harsh Raj **Cyber Job Simulation**

Certificate of Completion
September 20th, 2026

Over the period of September 2026, Harsh Raj has completed practical tasks in:
Cyber security



Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code 6aaffdfe875b6871a8c538f6 | User Verification Code 6aaffc4f8b93e7892a00f2fb | Issued by Forage

Figure: Deloitte Cyber Job Simulation – Certificate of Completion

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1. PROJECT DESCRIPTION

The **Deloitte Cyber Job Simulation** is a cybersecurity-focused virtual experience completed through the **Forge** platform. The certificate confirms that Harsh Raj completed practical tasks in Cyber Security during September 2026.

The main practical exercise involved investigating a simulated cybersecurity incident affecting **Daikibo Industrials**. A major news publication had revealed sensitive private information about the organisation while a production problem had caused its assembly lines to stop.

The client suspected that its newly introduced manufacturing status dashboard may have been compromised. The simulation required investigation of whether an external attacker could access the dashboard directly from the internet and which user ID demonstrated the most suspicious activity in the supplied web activity log.

2. INTRODUCTION

Cybersecurity involves protecting information systems, applications, networks and data from unauthorised access, misuse and disruption. An important part of cybersecurity is the ability to investigate incidents after suspicious activity has been detected.

Log analysis is one of the fundamental activities used during such investigations. Web activity logs can provide information about requests made to an application, including timestamps, HTTP methods, requested resources, response status codes and authenticated user identifiers.

The Deloitte Cyber Job Simulation provided practical exposure to this type of investigation through a simulated business scenario. The exercise centred on the telemetry/manufacturing status dashboard of Daikibo Industrials and required analysis of web activity to identify anomalous behaviour.

The exercise provided an opportunity to apply analytical thinking to cybersecurity evidence and understand how suspicious behaviour can be identified from seemingly ordinary web requests.

3. ORGANISATION & PLATFORM PROFILE

3.1 Deloitte

Deloitte was the organisation associated with the Cyber Job Simulation. The completion certificate identifies the experience as the **Deloitte Cyber Job Simulation** and records practical tasks in Cyber Security.

The simulation was designed around a realistic business cybersecurity scenario, allowing participants to work through investigation-oriented tasks. For this report, Deloitte is considered the organisation providing the simulated professional context rather than an employer or internship provider.

3.2 Forage

The simulation was completed through **Forage**, which issued the Certificate of Completion. The certificate records the completion date as **September 20, 2026** and includes verification information associated with the experience.

The virtual experience format allowed the participant to work through practical tasks representing aspects of cybersecurity work.

4. PROBLEM STATEMENT

Daikibo Industrials experienced a production problem that caused its assembly lines to stop. This disruption posed a potential risk to supply chains that relied on Daikibo's products.

At the same time, a major news publication revealed sensitive private information about the organisation. As a result, Daikibo suspected that the security of its new manufacturing/status dashboard may have been breached.

The cybersecurity investigation therefore needed to determine:

- Whether an external attacker could directly access the dashboard through the internet.
- Whether suspicious activity was present in the supplied web activity logs.
- Whether the request patterns differed from expected manual dashboard usage.
- Which user ID was associated with the suspicious activity.

5. PROJECT OBJECTIVES

- Assess possible dashboard accessibility from the internet.
- Understand the structure of the web activity logs provided for investigation.
- Identify normal web request sequences associated with dashboard usage.
- Detect suspicious request patterns within the available log data.
- Examine request timing to identify potentially automated behaviour.
- Trace suspicious requests to a user ID.
- Document the investigation findings in a structured manner.
- Develop practical cybersecurity investigation skills through a simulated professional scenario.

6. SCOPE OF THE JOB SIMULATION

The scope of this simulation was limited to the cybersecurity investigation activities provided within the Deloitte/Forage experience.

- Cybersecurity incident investigation.
- Web activity log analysis.
- Authentication and authorised-user activity.
- HTTP GET and POST requests.
- Dashboard resource requests.
- API request analysis.
- Request timestamps and frequency.
- Identification of unusual request sequences.
- Identification of suspicious user activity.

The exercise did not involve deploying security software, modifying a production system or conducting an attack against a real organisation. The work was performed within the simulated environment and using the supplied case material and log data.

7. TECHNOLOGIES, TOOLS & CONCEPTS

7.1 Web Activity Logs

The primary source of evidence was the **web_activity.log** file. The file contained activity organised into blocks based on unique internal IP addresses. Each block contained timestamps, HTTP methods, requested resources, response status and authorised user IDs where applicable.

7.2 HTTP Requests

- **GET** – used for requesting resources or information.
- **POST** – used in the login process.

7.3 Authentication

Authentication activity was important because requests following successful login contained an **authorizedUserId**. This identifier allowed activity to be associated with a particular authenticated user session.

7.4 API Requests

The dashboard generated API requests for factory and machine status. Examples included `/api/factory/status?factory=*` and `machine-status` endpoints. These requests were important because their frequency and timing helped reveal unusual behaviour.

7.5 Log Analysis

- Request order
- Request frequency
- Request timestamps
- Status codes
- User IDs
- Repeated API requests
- Length of request sequences

8. METHODOLOGY

The investigation followed a structured process, beginning with understanding the incident scenario and ending with documentation of the suspicious activity.

Step	Activity
1	Understand the Daikibo incident scenario
2	Determine whether direct internet access was possible
3	Understand the structure of the web activity log
4	Identify the normal login-to-dashboard request sequence
5	Examine longer request sequences
6	Analyse API request timing and repetition
7	Identify suspicious automated-looking activity
8	Trace the activity to the associated user ID
9	Document the findings

Investigation Workflow

Incident Scenario → Access Assessment → Log Examination → Normal Activity Identification → Suspicious Pattern Detection → User ID Identification → Findings & Documentation

9. WORK PERFORMED

9.1 Understanding Normal Dashboard Activity

The first stage was to understand how a normal user interacted with the dashboard. A typical session followed a recognisable sequence: initial request, login page, POST login request, dashboard page, CSS/JavaScript resources, factory-status API request and machine-status API requests.

This sequence was important because suspicious behaviour could then be compared against expected dashboard usage.

9.2 Examining Request Timing

The simulation highlighted the importance of checking whether automated API requests occurred at an exact interval. Normal human interaction with a dashboard would generally involve actions such as logging in, loading resources and requesting specific machine information. Repeated requests occurring at highly regular intervals can indicate automated behaviour and therefore warrant further investigation.

9.3 Examining Suspicious API Activity

The investigation identified a user ID associated with repeated machine-status API requests. The requests were made against multiple factories and repeatedly occurred at exactly one-hour intervals. The same user ID appeared repeatedly across these requests.

10. WEB ACTIVITY LOG ANALYSIS

10.1 Suspicious User ID

The user ID identified through the investigation was:

mdB7yD2dp1BFZPontHBQ1Z

The supplied log repeatedly associates this identifier with machine-status API requests across different factories.

10.2 Repeated One-Hour Pattern

One of the strongest indicators was the timing of the requests. The log shows activity at timestamps such as **20:00:48, 21:00:48, 22:00:48 and 23:00:48**, with the pattern continuing on the following day.

This exact repetition is significant because the task instructions stated that the dashboard did not continuously poll or push data and that users needed to refresh the page to obtain the latest information. Therefore, the repeated hourly requests represented an unusual pattern requiring investigation.

10.3 Multiple Factory Requests

The suspicious activity was not limited to a single machine or factory. The log shows repeated requests for machine status across multiple factories, including **Meiyo, Seiko, Shenzhen and Berlin**. The same authorised user ID was associated with these requests.

10.4 Authentication Status

Another notable feature was the appearance of **401 (UNAUTHORIZED)** responses during the repeated activity associated with this user ID. Later, a new login sequence appears, after which successful API requests resume. This illustrates how response status codes and authentication events can provide additional context during an investigation.

11. KEY FINDINGS

Finding 1 — No Direct Internet Access

The first task asked whether an attacker could access the manufacturing status dashboard directly from the internet. The answer in the simulation was: **“No, the attacker has no direct access to the status dashboard.”** The investigation therefore did not identify direct public-internet access to the dashboard as the access route considered in the exercise.

Finding 2 — Suspicious Automated-Looking Activity

The log analysis revealed a repeated request pattern associated with one user ID. The requests were made at highly regular intervals and targeted machine-status API endpoints across several factories.

Finding 3 — Suspicious User ID

The user ID associated with the suspicious activity was **mdB7yD2dp1BFZPontHBQ1Z**. The log repeatedly associates this identifier with the unusual machine-status request pattern.

Finding 4 — Unauthorised Responses

Repeated **401 (UNAUTHORIZED)** responses were also observed during the activity associated with this user ID. These responses provided additional evidence that the request sequence differed from a straightforward successful browsing session.

12. LEARNING OUTCOMES

- **Web Log Analysis:** Practical exposure to reading structured web activity logs and extracting information from timestamps, HTTP methods, requests and response codes.
- **Identifying Suspicious Behaviour:** Understanding that repetition, timing and unusually long request sequences can be important indicators.
- **Authentication Analysis:** Understanding how login activity and authorised user identifiers can connect requests to sessions.
- **API Activity Analysis:** Understanding how API requests can be examined to determine what information is being accessed and how frequently.
- **Analytical Thinking:** Comparing expected behaviour with observed behaviour rather than immediately assuming an unusual request was malicious.
- **Incident Investigation:** Exposure to a structured cybersecurity investigation process from incident understanding through evidence analysis.
- **Professional Cybersecurity Awareness:** Understanding how available evidence can support cybersecurity incident investigation.

13. CONCLUSION

The **Deloitte Cyber Job Simulation**, completed through Forage, provided practical exposure to cybersecurity investigation through a simulated business scenario.

The main case involved Daikibo Industrials, where sensitive information had been revealed while the organisation was experiencing a production disruption. The suspected compromise of the manufacturing status dashboard required an examination of access possibilities and web activity logs.

During the investigation, I studied the structure of the web activity log and identified the normal sequence of login, dashboard resource loading and API requests. I then examined longer request sequences and focused on the timing and repetition of machine-status API requests.

The investigation identified the user ID **mdB7yD2dp1BFZPontHBQ1Z** as being associated with the suspicious request pattern. The activity included repeated requests across multiple factories at highly regular one-hour intervals, along with repeated unauthorised responses.

The simulation also demonstrated the importance of understanding normal application behaviour before identifying anomalies. Overall, the experience strengthened my understanding of web logs, authentication activity, API requests, request timing and evidence-based cybersecurity investigation.

It provided a valuable practical foundation for further study and projects in the field of **Cyber Security**.

14. BIBLIOGRAPHY / REFERENCES

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